

Structure Learning of Agency: Recovering Interpretable Generative Models from Behaviour with Tensor Networks

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Abstract. Computational phenotyping of agency has two halves: fitting a generative model to behaviour, and reading agency-relevant quantities off the fitted model. Prior work established the second half on hand-specified models and left the first as a companion sketch. We complete the loop. A matrix product state (an interpretable tensor network) is trained unsupervised on action–observation rollouts of a T-maze partially observable Markov decision process, so its generative model $p(o, a)$ is learned from data rather than stipulated; we then recover the model’s structure in five senses that a bare density estimate does not provide. From the learned network we (i) extract normalised hidden states by a gauge-invariant, predictive-equivalence criterion—closing a gap left open by earlier tensor-network work, whose bond indices were “not normalised and therefore not usable”; (ii) reconstruct the labelled likelihood $A = p(o \mid s)$ and transition $B = p(s' \mid s, a)$ and match them to the ground-truth environment (emission total-variation 0.000, transition branching exact); (iii) recover the conditional-independence structure by exact mutual information; (iv) discover that the observation factorises, with context a near-independent modality; and (v) select the number of hidden states by a model-selection criterion rather than a truncation threshold, which we cast as active inference model expansion and Bayesian model reduction. On the learned model we then phenotype control capacity: empowerment reproduces the analytic zero-, intermediate-, and maximal-control regimes to three decimals, and epistemic action increases empowerment without it being an explicit objective. The pipeline is architecture-agnostic—the tensor network is not itself an active-inference agent—so the criteria and probe apply to any system whose sensorimotor stream can be modelled.

Keywords: Active inference · Agency · Computational phenotyping · Empowerment · Tensor networks · Structure learning · World models

1 Introduction

Agency in artificial systems is increasingly asserted and rarely measured. A companion line of work [1] argues that a minimal, operational notion of agency rests on three criteria—*intentionality* (action grounded in beliefs and desires), *rationality* (normatively coherent action entailed by a world model), and *explainability* (action causally traceable to internal states)—and that active inference [2] realises all three jointly in a partially observable Markov decision process (POMDP), whose likelihood and transition tensors constitute an inspectable world model. On that model, *empowerment*—the channel capacity between actions and future observations [3,4]—serves as an operational probe of the control capacity that intentional action presupposes.

That programme borrows its method from computational psychiatry’s *phenotyping* [5,6]: (i) fit a generative model to behavioural data, and (ii) explain variation from the inferred parameters. The worked example performed only the second half, on models specified by hand. This paper performs the first half, and shows that once it is done the second half returns not just a number but the entire interpretable model behind it.

Concretely, we train a *matrix product state* (MPS), a one-dimensional tensor network, as an unsupervised generative model of T-maze rollouts, following Wauthier et al. [7]. Tensor networks are attractive here because they admit exact marginalisation: the conditionals the empowerment probe needs, and the mutual informations a dependency analysis needs, are available in closed form from reduced density matrices. But a trained density model is not yet a *structured* one. Wauthier et al. observed that although the bonds between tensors “can be regarded as internal states, they are not normalised and therefore not usable,” and consequently bypassed hidden-state recovery entirely. Our central technical contribution is to supply exactly that missing piece and to build a full structural readout on top of it.

Contributions. From MPS models trained on the minimal four-observation T-maze and the full pymdp T-maze (24 composite observations), we recover, and validate against analytic ground truth:

1. **Hidden states** by gauge-invariant fidelity clustering, corrected to a *predictive-equivalence* criterion that is robust to gauge coherences the model writes into the bond (Sec. 3);
2. **A labelled generative model**: emission $A = p(o \mid s)$ and transition $B = p(s' \mid s, a)$, matched to the environment up to a state permutation (Sec. 4);
3. the **conditional-independence graph** over observables, by exact mutual information (Sec. 5);
4. the **observation factorisation**, from subsystem mutual information of the single-site reduced density matrix (Sec. 6);
5. the **number of hidden states** by model selection, framed as active inference model expansion and Bayesian model reduction (Sec. 7); and
6. **symbolic states**: a gauge that makes the bond index itself a labelled hidden state (Sec. 8).

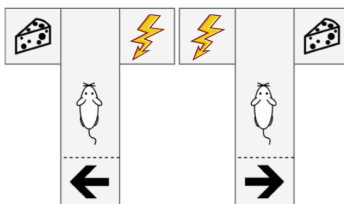


Fig. 1. The T-maze. The arrow marks the observation revealing where the cheese is; entering an arm is an absorbing trap, so the epistemically optimal first move is the cue.

On the learned model we then phenotype control capacity (Sec. 9). All code accompanies the paper.

2 Background

2.1 The task and the analytic model

The minimal T-maze is a two-step POMDP with three actions (Left, Right, Cue) and four observations (Cheese, Shock, Left, Right); cheese and shock are placed left/right with probability $1/2$ and the cue reveals their location, while entering an arm is an absorbing trap (Fig. 1) [1]. The full `pymdp` [8] T-maze factors the observation into Position (4), Reward (3), and Context (2), yielding 24 composite observations over a three-step episode beginning with a null action. Both admit an exact generative model $q(o, a)$, which we use only as ground truth for validation—the learner never sees it.

2.2 MPS as a generative model

An MPS represents the amplitude of a length- L sequence as a chain of contracted tensors, one per time step, each carrying a physical index (here the pair [action, observation]) and two bond indices. Trained by the Born rule $p(x) = |\psi(x)|^2/Z$ [9], it is a *Born machine*, and admits exact contraction of any marginal or conditional. Non-negative MPS are exactly hidden Markov models, with the bond index a classical hidden state [10]; a generic complex Born machine is strictly more expressive and has no such exact form, so hidden states must be recovered rather than read off (Sec. 3). We train each model with a DMRG-style cumulant gradient sweep and a learning-rate schedule that shrinks the step when the loss stalls [9]; the minimal model reaches $\|p - q\|_1 = 0.13$ and the full model $\|p - q\|_1 = 0.005$.

2.3 Empowerment

Empowerment at a state is the channel capacity between the next action and the next observation, $\max_{p(a)} I(A; O')$, computed by the Blahut–Arimoto algorithm [4]. Under the true environment conditional it is *objective* empowerment;

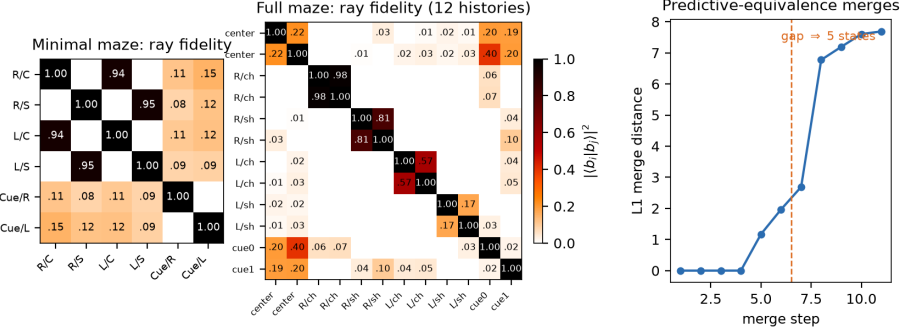


Fig. 2. State recovery. Pairwise bond-ray fidelity for the minimal maze (left, four blocks) and full maze (centre); raw fidelity over-separates on the full maze because the uninformative context bit is written into the bond. Right: predictive-equivalence merge distances—the near-zero merges collapse the gauge duplicates and the gap fixes the count at seven.

under the agent’s (here, the learned model’s) conditional it is *subjective* empowerment, and the two coincide as the fit improves. We report subjective potential empowerment over the full observation and over the reward modality alone (the preference-relevant variant).

3 Recovering hidden states

The bond of a trained MPS is defined only up to a gauge transformation $A^{(n)} \rightarrow A^{(n)}G$, $A^{(n+1)} \rightarrow G^{-1}A^{(n+1)}$, so the bare bond index has no intrinsic labelling. We first fix states in a gauge-invariant way: each action–observation history h induces a bond vector $|b_h\rangle$, and histories are clustered by pairwise fidelity $|\langle b_i | b_j \rangle|^2$ (Fig. 2, left/centre).

On the minimal maze this yields exactly four states (cheese-trap, shock-trap, and one per cue context), merging the two arms of entry because position is unobservable there. On the full maze, however, raw fidelity clustering *over-separates*: the model writes the uninformative context bit of the mid-step observation into the bond as a coherence that does not affect the future, splitting otherwise-identical states. The robust criterion is *predictive equivalence*—the predictive-state / observable-operator definition of state [11,12]: two histories are the same state iff they induce the same future conditional $p(o' | a', h)$. Clustering the action-conditioned predictive signatures, and cutting the dendrogram at its largest gap (Fig. 2, right), collapses the gauge duplicates and returns the correct seven states (unresolved start, four arm–outcome traps, two cue contexts). The learned state space thus tracks distinguishability, not surface history—arm-of-entry histories merge when position is hidden and separate when it is observed.

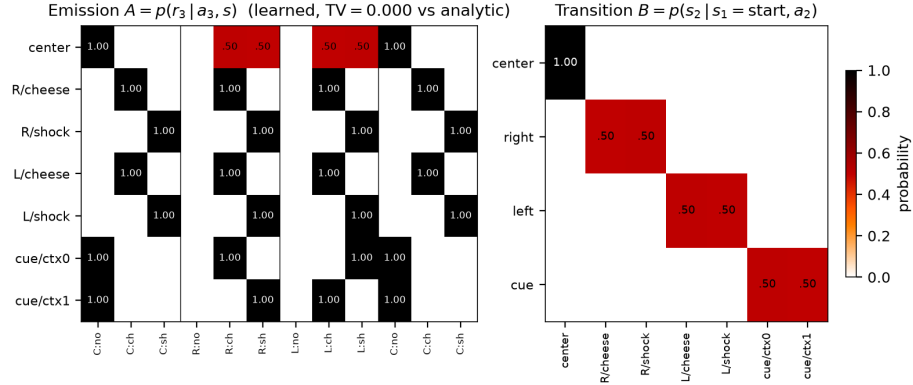


Fig. 3. Recovered generative model (full maze). Left: emission $A = p(r_3 | a_3, s)$ over seven states and four actions $\times$ three rewards; each state's pattern matches analytic ground truth (mean TV 0.000). Right: transition $B = p(s_2 | s_1 = \text{start}, a_2)$, recovering the exact 0.50/0.50 arm and cue branching.

4 A labelled generative model: emission and transition

Given the states, we reconstruct the factorised model. Because probabilities are Born-squared, we push amplitudes to probabilities *before* forming any stochastic object: attempting to read a stochastic matrix off the complex tensor, or to gauge-transform the model into a non-negative form, is either ill-defined or NP-hard for a genuine Born machine [10]. The **emission** $A = p(o | s)$ is estimated by conditioning the exact joint on each state's histories and marginalising the action; the **transition** $B = p(s' | s, a)$ by re-contracting the bond after appending (a, o) , classifying the resulting state by predictive equivalence, and accumulating Born weight. States are matched to the environment by the Hungarian algorithm, the only residual gauge freedom being a discrete permutation.

The reconstruction is essentially exact (Fig. 3, Table 1). On the minimal maze the emission matches ground truth to total-variation 0.000 for all four states. On the full maze, all seven emissions match to mean total-variation 0.000: the four traps are deterministic, and the two cue states carry the correct *opposite* action-conditioned reward patterns (under Right, context 0 yields cheese and context 1 yields shock). The recovered transition reproduces the branching structure of the environment exactly: from the start, Right and Left each split 0.50/0.50 over their two arm-outcome states, Cue splits 0.50/0.50 over the two context states, and staying at the centre is deterministic. This is the object Wauthier et al. could not extract: a normalised, labelled, ground-truth-comparable A and B read from a network trained on behaviour alone.

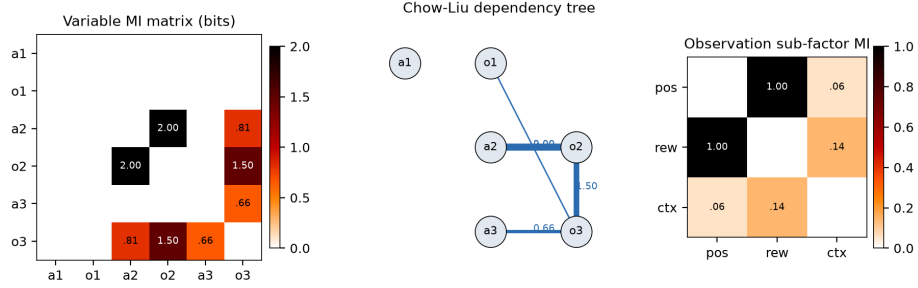


Fig. 4. Dependency and factorisation (full maze). Left: pairwise mutual-information matrix over the six observed variables. Centre: the Chow–Liu tree recovers the temporal action–observation chain. Right: subsystem mutual information of the single-site reduced density matrix—the model discovers context as a near-independent factor ($I(pos:ctx), I(rew:ctx) \approx 0.05$ bit) while position and reward are coupled (1.0 bit), the causally correct structure.

5 The dependency graph

Because the learned joint is available in closed form, its dependency structure follows from exact mutual information. We form the pairwise MI matrix over the observed variables and take its maximum-weight spanning tree (Chow–Liu [13]), the tensor-network analogue of reading correlation structure off a quantum state [14] (Fig. 4). The recovered tree is the correct temporal chain: the action at each step is strongly coupled to its observation and the observation to the next. Conditional mutual information supplies exact independence certificates: $I(a_1; o_3 | o_2) = 0$ (the first action affects the last observation only through the intervening one) and, in the minimal maze, $I(a_1; o_3 | o_2) = 0$ with $I(o_2; o_3 | a_2) = 1.31$ bits (dependence where the model should be dependent). We validate against the *observable-projected* ground-truth graph, since marginalising the latent state induces real dependencies among observables that a naive comparison to the full dynamic Bayes net would misread as error.

6 Discovering the observation factorisation

The full-maze model is fed a *flat* 24-dimensional observation; the Position×Reward×Context factorisation is assumed only in analysis. We test whether the learned tensors respect it, from the single-site reduced density matrix $\rho = \text{Tr}_{\text{rest}} |\psi\rangle\langle\psi|$. Subsystem quantum mutual information $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$ vanishes iff the factors are independent [15]. The model discovers context as a *near-independent* modality ($I(pos:ctx), I(rew:ctx) \approx 0.05$ bits) while position and reward remain coupled (1.0 bit)—which is causally correct, since position gates whether a reward is possible (Fig. 4, right). A search over groupings of the 24 indices confirms it: the trace-distance residual $\frac{1}{2} \|\rho - \bigotimes_k \rho_k\|_1$ is far lower for

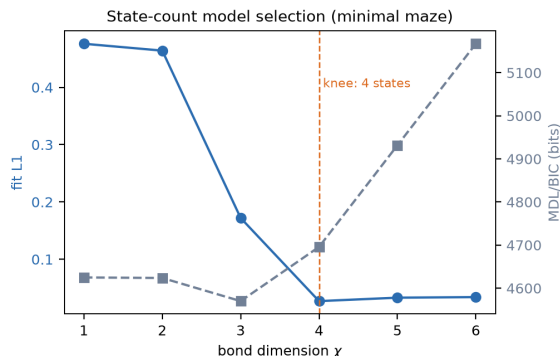


Fig. 5. State-count selection (minimal maze). Fit L1 (blue) drops $\sim 6\times$ from $\chi = 3$ to $\chi = 4$ and then plateaus; bond dimension beyond four is spare capacity. The description-length score (grey) is one state more conservative.

(position, reward) $\times$ context (0.21) than for full separation (0.54). The model organises the observation along the environment’s causal axes without being told they exist.

7 Selecting the number of states

That the recovered dimensionality is a *model-selection* result, not an artefact of an SVD threshold, is shown by sweeping the capped bond dimension χ and training each model to convergence (Fig. 5). On the minimal maze the fit improves sharply until $\chi = 4$ (fit L1 $0.17 \rightarrow 0.03$) and then goes flat: extra bond dimension is unused “spare capacity.” We read this through active inference structure learning [16,17]: a large- χ model has spare capacity; training commits capacity to states that raise model evidence (accuracy gain exceeding complexity cost) and leaves the rest unused, so the selected count is the fixed point of a free-energy trade-off over model structure—model expansion followed by Bayesian model reduction. A description-length penalty is one state more conservative here, because the fourth state adds interpretable low-probability structure that the fit needs but the likelihood barely rewards; the L_1 knee is the decisive signal.

8 Symbolic states by gauge fixing

Finally we remove the post-hoc character of state extraction by choosing the residual gauge that rotates the bond into the cluster basis. From the k cluster representatives we build an orthonormal frame (the eigenbasis of the history-aggregated bond density is more robust than a raw QR) and complete it to a unitary U ; applying U as the gauge preserves the represented distribution exactly (here to 4×10^{-14}) while the bond index becomes a labelled state that

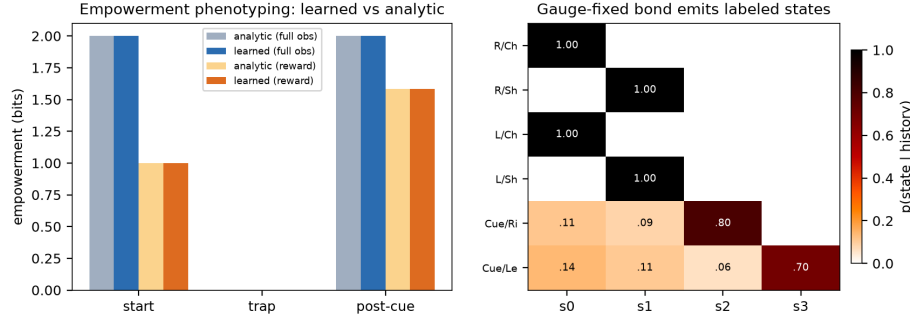


Fig. 6. Left: empowerment phenotyping on the learned full-maze model matches analytic across the zero-, intermediate-, and maximal-control regimes, full-observation and reward-only. Right: the gauge-fixed bond emits labelled states (minimal maze); trap histories are near one-hot, cue histories softer.

the model *emits*, $p_i(h) = |\langle q_i | U^\dagger | b_h \rangle|^2$ (Fig. 6, right) [18,19]. On the minimal maze the emitted distributions agree with the fidelity clustering (adjusted Rand index 1.0), near one-hot for the trap states and softer for the cue states—the expected signature of a genuinely complex Born machine, for which an exact classical-state gauge does not exist [10], quantified rather than hidden.

9 Phenotyping control capacity on the learned model

With the model recovered, we phenotype it. Subjective potential empowerment computed from the converged full-maze network reproduces the analytic regimes essentially exactly (Fig. 6, left; Table 1): 2.00 bits at the start, 0.00 in the trap, and 2.00 bits (full) / 1.585 bits (reward) post-cue, against analytic 2.00/1.585. The reward-modality restriction recovers the three control regimes of the minimal maze (1.00/0.00/1.585 bits), so restricting the probe to preferred modalities surfaces the agency-relevant structure that the full observation space obscures. The subjective–objective gap is now empirical and small: a bond-truncated checkpoint attenuates post-cue empowerment (0.65 bits reward-only), and it approaches the objective value as the fit improves. Epistemic action increases empowerment—the cue raises it from the intermediate to the maximal regime—without empowerment being any part of the training objective.

10 Discussion

What is new. Earlier tensor-network work learned $p(o, a)$ but could not expose normalised hidden states, and the phenotyping programme it was meant to serve was demonstrated only on hand-specified models. We supply the missing state construction—gauge-invariant and predictive-equivalence-based—and build on it a complete structural readout: a labelled likelihood and transition validated

Table 1. Structure recovery against analytic ground truth. Emission/transition are total-variation distances; empowerment is bits (full/reward). “Learned” is the converged full-maze MPS ($\|p - q\|_1 = 0.005$).

Quantity	Analytic	Learned	Error
Emission $A = p(o \mid s)$ (mean TV over 7 states)	—	—	0.000
Transition B branching (Right $\rightarrow$ {cheese, shock})	0.50/0.50	0.51/0.49	≤ 0.01
Empowerment, start	2.00/1.00	2.00/1.00	0.00
Empowerment, trap	0.00/0.00	0.00/0.00	0.00
Empowerment, post-cue	2.00/1.585	2.00/1.585	< 0.001
Hidden-state count (minimal / full)	4 / 7	4 / 7	exact

to $\sim 10^{-3}$, the conditional-independence graph, the observation factorisation, a principled state count, and a symbolic-state gauge. Agency phenotyping then runs on the learned model and matches the analytic regimes. The pipeline is architecture-agnostic: the tensor network is not an active-inference agent, so recovering active-inference-style structure from it addresses the circularity worry that active inference might satisfy the agency criteria only by construction.

Senses of structure learning. We are precise about the phrase. The model performs unsupervised discovery of latent dimensionality (Sec. 7), of the task-relevant state abstraction (Sec. 3), of the dependency graph (Sec. 5), and of the observation factorisation (Sec. 6), and it yields a labelled factored model (Sec. 4). It does not search over network topology; the tensor-network structure-search literature [20,21] would be the route to that, and is a natural next step on richer environments.

Limitations. Results are on two small POMDPs with known ground truth, so validation is against an analytic model rather than blind; a state-count criterion stated before inspection, and a held-out protocol on data an exact model cannot memorise, would strengthen the claim. Empowerment measures controllability, not agency: reading control regimes as agency profiles is licensed only because the agentic action chain is independently inspectable in the recovered model [1]. Finally, the description-length criterion for state count is conservative under our naive parameter count, which ignores gauge freedom; a gauge-aware count is future work.

Governance. Because the readout is architecture-agnostic and needs only a learnable model of the sensorimotor stream, it offers a concrete route to auditing opaque systems: fit an interpretable surrogate, recover its states and its controllability profile, and test whether governance handles track the recovered regimes [1]. Whether real interventions map onto these regimes is the empirical programme this pipeline makes testable.

11 Conclusion

Computational phenotyping of agency can be carried out end-to-end: from behaviour to a learned generative model, from that model to its recovered structure, and from the structure to a controllability phenotype that reproduces ground truth. The tensor network supplies exact conditionals and, once gauge-invariant states are fixed, a fully interpretable factored model—the inspectable surrogate that phenotyping of opaque systems requires.

Data and code. All code, models, and figures accompany the paper at <https://github.com/philipollaswilson/Tensor-Networks-for-the-Tmaze>.

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Disclosure of Interests. The authors declare no competing interests.

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